

PSEUDOCODE

START

DEFINE Structure Node

data

next

prev

END Structure

FUNCTIONS createNode (data)

Allocate memory for a new NODE

IF memory allocation fails THEN

Display "Memory allocation failed"

EXIT

END IF

newNode . data ← data

newNode . next ← NULL

newNode . prev ← NULL

RETURN newNode

END FUNCTION

FUNCTION printlist (head)

1) Insertion in singly linked list

PSEUDOCODE

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START

Create new node

IF position = beginning

new.next $\leftarrow$ head

head $\leftarrow$ new

ELSE

Traverse to required position

new.next $\leftarrow$ temp.next

END IF

Display list

STOP

2) Deletion in singly linked list

START

IF list is empty

Display "Empty"

ELSE IF beginning

head $\leftarrow$ head.next

ELSE

Traverse to required node

temp.next $\leftarrow$ temp.next.next

END IF

Display list

STOP

3) Searching

START

temp $\leftarrow$ head

position $\leftarrow$ 1

WHILE temp $\neq$ NULL

IF temp.data = key

Display position

STOP

temp $\leftarrow$ temp.next

position $\leftarrow$ position + 1

END WHILE

DISPLAY "NOT FOUND"

STOP

4) Reverse linked list

START

prev $\leftarrow$ NULL

current ← head

WHILE current ≠ NULL

next ← current.next

current.next ← prev

prev ← current

current ← next

Display list

STOP

5.) Count Nodes

START

count ← 0

temp ← head

while temp ≠ NULL

count ← count + 1

temp ← temp.next

END WHILE

DISPLAY count

STOP

key	locate	data	next
69	1	1	5
58	2	2	8
49	3	3	8
18	4	4	8
89	5	5	1